

Closed forms for the constants in the odd-weight identities between dihedral modular graph functions

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Abstract We consider the combinations X_w of two-loop dihedral modular graph functions $C(a, b, c)$ of odd weight w whose hyperbolic Laplacian is a multiple of the Eisenstein series $E(w)$. At each odd weight examined, $3 \leq w \leq 61$, we prove $X_w = f_w E(w) + g_w \zeta(w)$ with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$. The proof combines exact Laplace algebra, a harmonicity, invariance and growth argument, and a constant term evaluated from a Laurent polynomial in rational arithmetic. It assumes a theorem of D'Hoker–Kaidi and a standard lemma, neither checked by code. We also show exactly that at each weight $3 \leq w \leq 25$ the space of combinations with $\Delta X \in \mathbb{Q}E(w)$ has dimension 1 for odd w and 0 for even w . Independently of the proof, the constants at $w = 3, \dots, 17$ are confirmed numerically at 32 digits at verifier-chosen points, and $w = 17$ was a preregistered blind test. The full weight-11 basis has exactly one relation, established numerically by a singular-value criterion. A proof beyond the examined range remains open.

Keywords Modular graph functions · Laplace eigenvalue equations · Eisenstein series · Bernoulli numbers · Genus-one string amplitudes · Computer-assisted proof

1 Introduction

Two-loop dihedral modular graph functions $C(a, b, c)$ appear as coefficients in the low-energy expansion of the one-loop amplitude of closed strings. D'Hoker, Green and Vanhove showed that they satisfy Laplace eigenvalue equations with inhomogeneous terms polynomial in non-holomorphic Eisenstein series, and that the Laplace equation fixes an algebraic identity only up to an additive constant of integration [1]. The DGV combination X_w of $C(a, b, c)$, whose Laplacian is proportional to $E(w)$, is used here at each odd weight examined, $3 \leq w \leq 61$. The constants are explicit in that work only up to weight 9, and g_7, g_9 appear there as undetermined integration constants. The constant g_7 was later given in a general procedure based on Laurent polynomials [2]. Identities among two-loop graphs at all weights were also constructed earlier [3], and relations between graphs with four and five links have been proved [4].

Open question. What are the closed forms of f_w and g_w in $X_w = f_w E(w) + g_w \zeta(w)$ for odd w ?

Difficulty. The Laplace equation alone leaves g_w undetermined [1]. Fixing it requires the Laurent polynomial of each $C(a, b, c)$ at the cusp. The size of the combination grows with w , and numerical searches for relations are limited by the working precision relative to the height of the coefficients.

Contribution. The Laurent polynomial of each single $C_{u,v,w}$ is known (Theorem 5.1 of D'Hoker–Kaidi, arXiv:1902.04180). We evaluate it in rational arithmetic on the DGV combinations. This gives closed forms for f_w and g_w with a proof at each odd weight examined, $w \leq 61$, together with numerical confirmations independent of that proof.

Summary of results.

- Theorem 1 shows $X_w = f_w E(w) + g_w \zeta(w)$ exactly at each odd weight examined, $3 \leq w \leq 61$, with f_w and g_w in closed form. The proof is computer-assisted in rational arithmetic, and its assumptions are stated openly.
- Theorem 2 shows that the space $\{X : \Delta X \in \mathbb{Q}E(w)\}$ is one-dimensional for odd w and zero for even w , at each weight $3 \leq w \leq 25$.
- Conjecture 1 states that the closed forms persist beyond the examined range; they are proved up to $w = 61$.
- Observations 1 to 3 confirm numerically, independently of the proof, the constants at $w = 3, \dots, 17$. They show that the full weight-11 basis carries exactly one relation. They also report a hardening run that exposed and fixed one evaluator bug.

2 Setting

Let $\tau = \tau_1 + i\tau_2$ and $p = m\tau + n$ with $(m, n) \in \mathbb{Z}^2 \setminus \{0\}$. The non-holomorphic Eisenstein series and the dihedral two-loop graphs are

$$E(s) = \sum_p' \left(\frac{\tau_2}{\pi|p|^2} \right)^s,$$

$$C(a, b, c) = \sum_{p_1+p_2+p_3=0} \prod_{i=1}^3 \left(\frac{\tau_2}{\pi|p_i|^2} \right)^{a_i},$$

with $(a_1, a_2, a_3) = (a, b, c)$ and weight $w = a + b + c$. The hyperbolic Laplacian is

$$\Delta = \tau_2^2 (\partial_{\tau_1}^2 + \partial_{\tau_2}^2),$$

and $E(w)$ satisfies

$$\Delta E(w) = w(w-1)E(w).$$

ζ is the Riemann zeta function and B_k are Bernoulli numbers. For odd $w = 2\mu + 3$, X_w denotes the rational combination of $C(a, b, c)$ of weight w given by eq. 3.57 of D'Hoker–Green–Vanhove (arXiv:1502.06698), in that normalisation. The constants are

$$f_w = \frac{3((w-1)/2)!}{w},$$

$$g_w = \frac{6|B_{w-1}|}{((w-1)/2)!}.$$

Holomorphic Eisenstein series are normalised by $a_0 = 1$, $E_k = 1 - \frac{2k}{B_k} \sum_n \sigma_{k-1}(n)q^n$ with $q = e^{2\pi i\tau}$.

Verifier parameters. The numerical verifier uses 4 evaluation points with seed 4711 in the region $|\tau_1| \leq 1/2$, $1 \leq \tau_2 \leq 2.5$. The working precision is 32 digits. Numerical relations are accepted at maximal relative residual $1e-24$, and at $1e-14$ when a Laplacian is involved. The Laplacian is evaluated by a fourth-order finite-difference scheme with step $h = 1e-6$.

3 Closed forms and their proof

Theorem 1 (Closed forms for odd weight). *At each odd weight w examined, $3 \leq w \leq 61$, one has $X_w = f_w E(w) + g_w \zeta(w)$, with the constants*

$$f_w = \frac{3((w-1)/2)!}{w}$$

$$g_w = \frac{6|B_{w-1}|}{((w-1)/2)!}$$

Proof. The proof has three steps, following.

1. Exact rational computation in the algebraic Laplace representation gives $\Delta X_w = w(w-1)f_w E(w)$ with no other terms, re-done at each odd weight examined, $w \leq 61$. Since $\Delta E(w) = w(w-1)E(w)$, the difference $X_w - f_w E(w)$ is harmonic.
2. The difference is $SL(2, \mathbb{Z})$ -invariant and of polynomial growth at the cusp, hence constant by the standard argument used by D'Hoker–Green–Vanhove.
3. The constant equals the τ_2^0 term of the Laurent polynomial of X_w . This polynomial follows from Proposition 2.1 and Theorem 5.1 of D'Hoker–Kaidi (arXiv:1902.04180), evaluated in rational arithmetic. The transcription was checked against their eq. 5.19, and every Laurent term not involving products of zeta values agrees with that of $f_w E(w) + g_w \zeta(w)$. This is a consistency check. The τ_2^0 term equals $g_w \zeta(w)$ with $g_w = 6|B_{w-1}|/((w-1)/2)!$ at each odd weight examined, $w \leq 61$.

Assumptions not checked by code: the correctness of Theorem 5.1 of arXiv:1902.04180, proved there in Appendix A, whose conjectural part concerns only the coefficient of τ_2^{2-w} and is not used, and the standard lemma in step 2. $\square$

Table 1 Constants f_w and g_w in $X_w = f_w E(w) + g_w \zeta(w)$ for odd $w \leq 25$.

w	f_w	g_w
3	1	1
5	6/5	1/10
7	18/7	1/42
9	8	1/120
11	360/11	1/264
13	2160/13	691/327600
15	1008	1/720
17	120960/17	3617/3427200
19	1088640/19	43867/48263040
21	518400	174611/199584000
23	119750400/23	77683/83462400
25	57480192	236364091/217945728000

Novelty status: partly known. The combinations X_w (DGV eq. 3.57), f_w up to weight 9 and g_w at weights 3 and 5 (DGV eq. 3.34) are due to DGV, and g_7 also appears in arXiv:2109.05017. Not given there are the closed forms for general odd w and their proof for $w \geq 9$. The Laurent polynomial of each single $C_{u,v,w}$ is used here as a tool.

Table 1 lists the values for odd $w \leq 25$.

Theorem 2 (Uniqueness of the harmonic combination). *At each weight w examined, $3 \leq w \leq 25$, the rational combinations X of dihedral $C(a, b, c)$ of weight w with $\Delta X \in \mathbb{Q} E(w)$ form a vector space of dimension 1 for odd w and 0 for even w . For odd w it is spanned by the DGV combination X_w , and $\Delta X_w = w(w-1)f_w E(w)$.*

Certificate. The algebraic Laplace representation of D'Hoker–Green–Vanhove was re-derived and the linear conditions were solved in exact rational arithmetic for each w in the range. $\square$

Novelty status: partly known. DGV state that the eigenvalue-0 eigenspace has multiplicity one for each odd weight and zero for each even weight; the closed form $f_w = 3((w-1)/2)!/w$ is not given there, and the exact confirmation up to $w = 25$ is an independent recomputation.

Remark 1 (Conjecture 1). The closed forms f_w and g_w of Theorem 1 are conjectured to persist at odd weights beyond the examined range. Before the proof, the formula was a conjecture read off from $w = 3..13$. It is now proved up to $w = 61$. A proof beyond that range is open.

Remark 2 (Status of the computer-assisted parts). The trusted base consists of exact rational arithmetic in the Laplace algebra and in the Laurent polynomials of Theorem 5.1 of arXiv:1902.04180, plus the two assumptions listed in the proof of Theorem 1. In the numerical observations the base is 32-digit arithmetic at verifier-chosen points, together with an exact check of the leading Laurent coefficient. A third party re-runs every certificate with ‘python -m asd.recheck modular’ from the repository given in Appendix A.

4 Exact Laplace computations at weights 9 and 11

Proposition 1 (Weight 9). *Exactly, in rational arithmetic,*

$$\Delta(9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3)) = 288 E(9)$$

so $9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3) - 4E(9)$ is annihilated by Δ .

Certificate. The algebraic Laplace representation of DGV was re-derived and checked in rational arithmetic; the verifier returns $\lambda = 288$. $\square$

Novelty status: already known for weights 3, 5, 7 and 9 (DGV eq. 3.33).

Proposition 2 (Weight 11). *Exactly, in rational arithmetic,*

$$\Delta(2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3)) = 100 E(11)$$

so $2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3) - \frac{10}{11}E(11)$ is annihilated by Δ .

Certificate. As for the weight-9 proposition; the verifier returns $\lambda = 100$. $\square$

Novelty status: the combination for each odd w is already known (DGV eq. 3.57); f_w and g_w are not given there.

5 Numerical confirmation, independent of the proof

The following observations are numerical. Each relation was checked at 4 verifier-chosen points (seed 4711, 32 digits). The leading Laurent coefficient was checked exactly and vanishes in every case.

Numerical observation 1 (Relations at $w = 3, \dots, 17$). As functions of τ , at relative residual below $1e-24$:

$$C(1, 1, 1) - E(3) - \zeta(3) = 0$$

$$30 C(2, 2, 1) - 12 E(5) - \zeta(5) = 0$$

$$252 C(3, 3, 1) + 252 C(3, 2, 2) = 108 E(7) + \zeta(7)$$

At weight 9,

$$\begin{aligned} & -2160 C(4, 4, 1) - 4320 C(4, 3, 2) \\ & -960 C(3, 3, 3) + 960 E(9) + \zeta(9) = 0 \end{aligned}$$

and at weight 11,

$$\begin{aligned} & 19008 C(5, 5, 1) + 38016 C(5, 4, 2) \\ & + 19008 C(5, 3, 3) + 28512 C(4, 4, 3) \\ & - 8640 E(11) - \zeta(11) = 0 \end{aligned}$$

Analogous relations hold at $w = 13$ (with $-691\zeta(13)$), $w = 15$ (with $-\zeta(15)$) and $w = 17$ (with $-3617\zeta(17)$).

Table 2 Maximal relative residuals of the weight- w relations at the 4 verifier points (bound $1e-24$).

w	max. rel. residual
3	5.13e-44
5	1.57e-43
7	1.59e-43
9	8.33e-44
11	1.21e-43
13	5.67e-44
15	6.46e-44
17	5.49e-44

Certificate. Per relation: exact leading Laurent coefficient 0, then numerical evaluation at the 4 points. The maximal relative residuals are in Table 2. The weight $w = 15$ was a prediction made before its value was read, and $w = 17$ a preregistered blind test. $\square$

Novelty status: weights 3 and 5 are already known (DGV eq. 1.7). Weight 7 is already known, since DKS eq. 3.77 gives $C(3, 3, 1) + C(3, 2, 2) = \frac{3}{7}E(7) + \zeta(7)/252$, i.e. f_7 and g_7 . For $w = 9$ the identity with an integration constant g_9 is in DGV eq. 3.34, but the value of g_9 is not given there. For $w = 11$ the constants for $w \geq 11$ are not given in DGV.

Numerical observation 2 (Full weight-11 basis). For the full weight-11 basis $C(9, 1, 1), \dots, C(4, 4, 3), E(11)$ and $\zeta(11)$, the rational linear relations form a space of dimension exactly 1, spanned by the weight-11 relation of the previous observation.

Certificate. The preregistered criterion 1 (15 points, seed 4712, all other singular values above $1e-8$) was **not** met: the smallest of the other singular values is $4.149e-9$, while the kernel singular value is $7.591e-44$. This is reported as a failure. Criterion 2 was preregistered afterwards on new data (20 points, seed 4713, $\tau_2 \in [0.9, 3]$) with threshold $1e-16$, chosen to lie well above the kernel value and well below the smallest non-kernel value, and it passed. The threshold separates the kernel direction (largest small singular value $1.11e-43$) from the smallest non-kernel value $3.69e-9$ by many orders of magnitude, which is why it distinguishes the two groups unambiguously. The 20×12 matrix has one singular value below $1e-24$ and 11 above $1e-16$. The singular-value gap supports completeness numerically; it is not a proof. $\square$

Novelty status: partly known; the constants for $w \geq 11$ are not given in DGV.

Numerical observation 3 (Hardening of the weight-11 identity). The weight-11 identity was tested at 8 points not used before (seed 9001, including τ near ρ and τ_2 up to 3), preregistered with 8 points planned and 8 completed. With the residual required to be at most 10^{-d} at working precision d , this holds at 8 of 8 points. An independent check by direct lattice summation in double precision agrees to at most $1e-12$ relative at 8 of 8 points.

Certificate. The run exposed and fixed an evaluator bug. The Fourier sum for E_s stopped when a term vanished, which happens at $\tau_1 = 1/4$, where $\cos(2\pi N\tau_1) = 0$ for odd N . Before the fix that point showed a precision-independent residual of $3.5\text{e-}10$. All listed points were computed with the corrected code, and no earlier test point was affected. $\square$

6 Discussion

Interpretation. Theorem 1 reduces the constants of the odd-weight identities to the single Laurent coefficient $g_w\zeta(w)$, which has the Bernoulli form $6|B_{w-1}|/((w-1)/2)!$ up to $w = 61$. The numerical confirmations at $w = 3..17$ test the identities directly as functions of τ , without the Laurent-polynomial input.

Limitations. The proof rests on Theorem 5.1 of arXiv:1902.04180 and on the standard lemma, neither re-proved here. Its range ends at $w = 61$. The relation-space statement for the full weight-11 basis is numerical, and its completeness criterion is a singular-value gap. The harmonic-space dimensions are exact only for $w \leq 25$.

What did not work. (a) The originally preregistered completeness criterion 1 for the full weight-11 basis failed, because the smallest non-kernel singular value $4.149\text{e-}9$ lies below the threshold $1\text{e-}8$; a second criterion, preregistered afterwards on new data, passed. (b) The hardening run found a premature termination of the Fourier sum for E_s at $\tau_1 = 1/4$, now fixed. Further negative results are collected in Appendix A.

Open questions.

- (i) Prove the identities beyond the examined range, presumably by reducing the Laurent constant to a binomial-sum identity.
- (ii) Determine the constants of the non-harmonic eigenvalue equations.
- (iii) Extend the analysis to higher loop order.

Declarations

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Code and data availability Repository <https://github.com/NinjaTurtlesHackathons/Hacknation>, branch `claude/modulformen-paper`, directory `projects/modular`. All certificates are reproduced with `python -m asd.recheck modular`, and the paper is rebuilt with `python -m asd.paper --domain modular`.

Competing interests None.

A Numerical methods and error control

Certificate types. (1) *Exact Laplace algebra:* rational arithmetic in the algebraic Laplace representation, as

Table 3 Relation-space check at weights 3, 5, 7 and 9: matrix size, relations confirmed, smallest large singular value.

w	matrix	dim	smallest large s.v.
3	6x3	1	0.731
5	7x4	1	0.0372
7	9x6	1	0.000127
9	12x9	1	8.6e-7

in Theorem 2 and the two propositions. (2) *Laurent constants:* rational arithmetic from Proposition 2.1 and Theorem 5.1 of arXiv:1902.04180, as in Theorem 1. (3) *Numerical relation:* an exact check of the leading Laurent coefficient in y , then evaluation at the verifier points (4 points, seed 4711, 32 digits). The tolerance is $1\text{e-}24$, or $1\text{e-}14$ with a Laplacian evaluated by finite differences with $h = 1\text{e-}6$. (4) *Relation space:* singular values of the normalised value matrix at seed 4712, plus individual confirmation of each relation. (5) *Sturm certificates* for holomorphic identities.

Trusted base. The trusted base is exact rational arithmetic, 32-digit arithmetic at the verifier’s own points, and the two uncoded assumptions of Theorem 1.

Verifier self-test. Adversarial variants with falsified constants were submitted to the verifier. Examples are a doubled $\zeta(17)$ term, which gives relative residual $3.78\text{e-}02$, and a doubled $\zeta(11)$ term, which gives $2.85\text{e-}01$. A wrong g_5 and a wrong λ in the exact Laplace algebra were likewise rejected. In total 53 counter-checks were run: 4 passed (consistency checks that do not contradict the statements), 48 did not pass, and 1 was not executable.

Relation spaces at weights 3 to 9. For the weight- w bases the relation space has dimension exactly 1, with a singular-value gap in the normalised value matrix (seed 4712).

Laplace equations (observations). At 4 verifier points (relative residual at most $1\text{e-}14$):

$$L[C(2, 1, 1)] - 2C(2, 1, 1) - 9E(4) + E(2)^2 = 0$$

$$L[C(2, 2, 1)] - 8E(5) = 0$$

$$L[C(2, 2, 1)] - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$$

$$L[C(3, 1, 1)] - 6C(3, 1, 1) - \frac{86}{5}E(5) + 4E(2)E(3) - \frac{1}{10}\zeta(5) = 0$$

Equations of this type are already stated in the literature [1], [3], [4].

Holomorphic validation (Sturm certificates). All identities are exact on the stated group, with all terms modular of the same weight and character. Each is certified by the vanishing of the q -coefficients up to the Sturm bound.

Table 4 Sturm-certified identities used as validation of the verifier.

identity	group	weight	bound
$\theta(1)^4 = \eta(2)^{20}\eta(1)^{-8}\eta(4)^{-8}$	$\Gamma_0(4)$	2	1
$\theta(1)^2 = \eta(2)^{10}\eta(1)^{-4}\eta(4)^{-4}$	$\Gamma_0(4)$	1	1/2
$\text{thetaE8} = E_4$	$\Gamma_0(1)$	4	1/3
$\text{thetaE8}^2 = \text{thetaD16}$	$\Gamma_0(1)$	8	2/3
$\text{thetaE8}^2 = E_4^2$	$\Gamma_0(1)$	8	2/3
$\text{thetaE8}^2 = E_8$	$\Gamma_0(1)$	8	2/3
$1728\Delta = E_4^3 - E_6^2$	$\Gamma_0(1)$	12	1
$j\Delta = E_4^3$	$\Gamma_0(1)$	12	1

The $\Gamma_0(4)$ identities are stated in the literature [5]. The expression of Δ through E_4, E_6 is also stated in the literature [6]. The `thetaE8`, `thetaD16` and `j` identities are reproductions of classical results, preregistered as validation anchors.

Theorem 3 (Eta-quotient span). *For all $N \leq 30$, holomorphic eta quotients with trivial character span $M_4(\Gamma_0(N))$ exactly for $N \in \{2, 4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, 24, 25, 26, 27, 28, 30\}$, and do not for $N \in \{1, 3, 7, 11, 13, 17, 19, 21, 23, 29\}$.*

Certificate. Exact and exhaustive computation of the span for each $N \leq 30$. *Novelty status:* not found in a targeted search of 33 abstracts on 2026-10-04.

Negative results.

- *Weight 6.* A search for rational linear relations among $C(4, 1, 1)$, $C(3, 2, 1)$, $C(2, 2, 2)$ and products of Eisenstein series and ζ values passed no claim at the verifier. The value matrix is 12x9, with 8 singular values above $1e-8$ and the smallest large one $2.44e-8$.
- *Eta quotients.* The first attempts at the table for $M_4(\Gamma_0(N))$ passed no claim; the table above is the later exact computation.
- *Weight 11.* A numerical integer-relation search over the full 12-element basis, with 6-8 points and up to 64 digits, found no relation. The run was stopped for lack of time, and the presumed cause is working precision that is too low relative to the coefficient height. The verified claim is the later criterion-2 result.
- *Weights 13 and 15.* First attempts to check the constant formula passed no claim at the verifier. The direct relation checks are in Table 2.

Laboratory workflow. The laboratory ran 17 rounds with 37 verified statements and 4 negative results. Every round was preregistered before the experiment. The agent roles recorded in the negative results are *sparsam*, *numeriker*, *skeptiker* and *theoretiker*. Counter-checks were generated adversarially against each answer, with statistics given above. Novelty searches were run for each result over the abstracts of the queried literature.

B Provenance of the statements

The provenance table mapping every statement to its evidence follows; it is generated automatically.

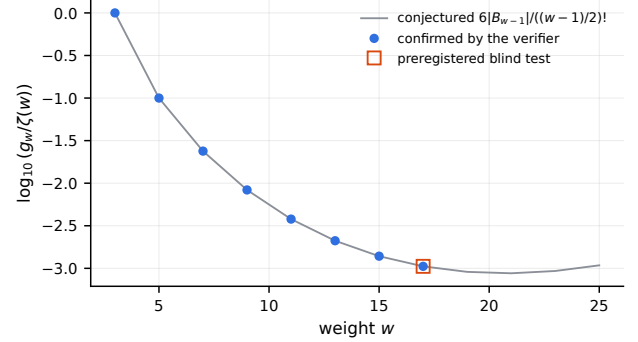


Fig. 1 Integration constants g_w of the odd-weight identities in the normalisation of eq. (3.57) of D’Hoker-Green-Vanhove: conjectured closed form (line) and the weights at which the verifier confirmed the identity numerically (dots, 32 digits, 4 verifier points); the square marks the preregistered blind test at $w = 17$.

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

References

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Statement	Level
Section: Introduction	computed_rigorous, observed
Section: Setting	computed_rigorous
Theorem 1 (Closed forms for odd weight)	computed_rigorous
Section: Closed forms and their proof	computed_rigorous
Theorem 2 (Uniqueness of the harmonic combination)	computed_rigorous
Remark 1 (Conjecture 1)	computed_rigorous
Remark 2 (Status of the computer-assisted parts)	computed_rigorous, observed
Proposition 1 (Weight 9)	computed_rigorous
Section: Exact Laplace computations at weights 9 and 11	computed_rigorous
Proposition 2 (Weight 11)	computed_rigorous
Section: Numerical confirmation, independent of the proof	computed_rigorous, observed
Observation 1 (Relations at $w = 3, \dots, 17$)	observed
Observation 2 (Full weight-11 basis)	observed
Observation 3 (Hardening of the weight-11 identity)	observed
Section: Discussion	computed_rigorous, observed
Section: Declarations	observed
Section: Appendix A: Numerical methods and error control	computed_rigorous, observed
Theorem 3 (Eta-quotient span)	computed_rigorous

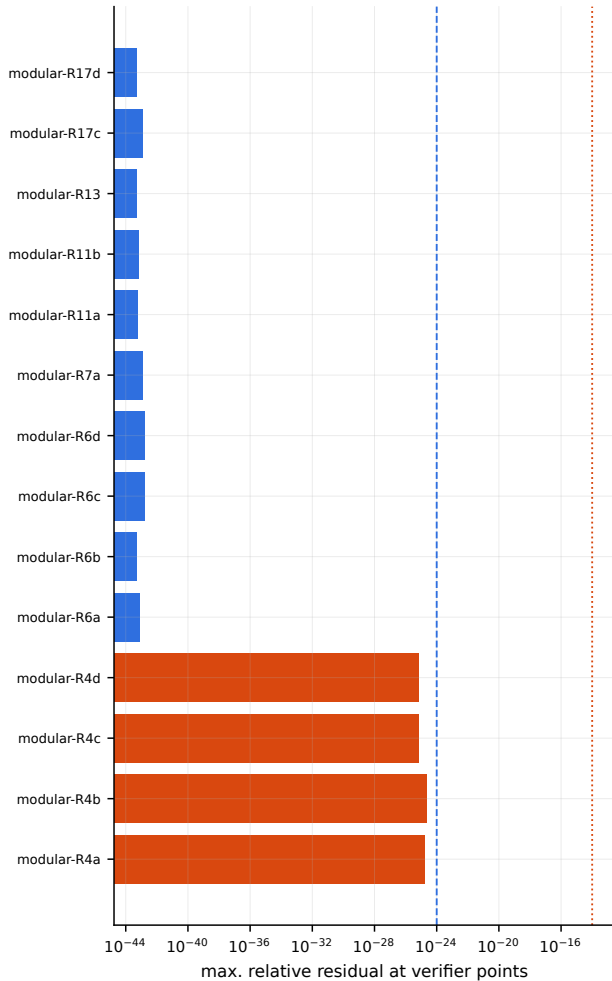


Fig. 2 Numerical verification margins: largest relative residual of each confirmed relation among modular graph functions at the verifier's own random points (32 digits). Dashed: acceptance threshold 10^{-24} ; dotted: threshold 10^{-14} for relations containing a Laplacian (orange).

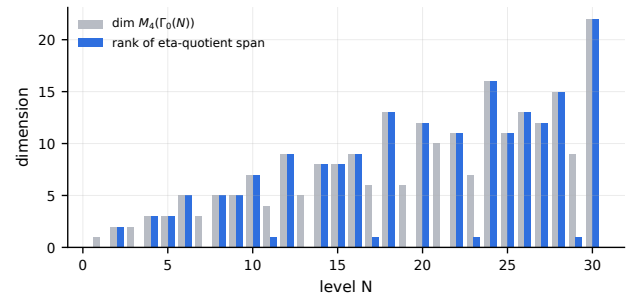


Fig. 3 Exact comparison, for every level $N \leq 30$, of $\dim M_4(\Gamma_0(N))$ (grey) with the rank of the span of all holomorphic eta quotients of weight 4 with trivial character (blue).